12 Triple Integrals

12.1 Interpretations of Triple Integrals

Let $\mathcal{W} \subseteq \mathbb{R}^3$ be a solid region and let $f(x, y, z)$ be integrable on $\mathcal{W}$. The triple integral

$$\iiint_{\mathcal{W}} f(x, y, z)\, \mathrm{d}V$$

has several important interpretations:

1. If f is a density function with units of amount per unit volume, then

$$\iiint_{\mathcal{W}} f(x, y, z)\, \mathrm{d}V$$

   is the total amount contained in $\mathcal{W}$. In particular, if $f = \rho$ is a mass density, then the integral gives the mass of the solid.

2. Taking $f = 1$ gives the volume of the region:

$$\mathrm{Vol}(\mathcal{W}) = \iiint_{\mathcal{W}} 1\, \mathrm{d}V.$$

3. The average value of f over $\mathcal{W}$ is

$$f_{\mathrm{avg}} = \frac{1}{\mathrm{Vol}(\mathcal{W})} \iiint_{\mathcal{W}} f(x, y, z)\, \mathrm{d}V.$$

4. Geometrically,

$$\iiint_{\mathcal{W}} f(x, y, z)\, \mathrm{d}V$$

   can be viewed as the signed hypervolume of the four-dimensional region determined by the graph of f over $\mathcal{W}$.

12.2 Evaluating Triple Integrals

As with double integrals, a triple integral may be approximated by a Riemann sum or evaluated exactly by writing it as an iterated integral.

Theorem 12.1: Fubini's Theorem for Triple Integrals

If f is continuous on the box

$$\mathcal{B} = [a, b] \times [c, d] \times [p, q],$$

then

$$\iiint_{\mathcal{B}} f(x, y, z)\, \mathrm{d}V = \int_a^b \int_c^d \int_p^q f(x, y, z)\, \mathrm{d}z\, \mathrm{d}y\, \mathrm{d}x.$$

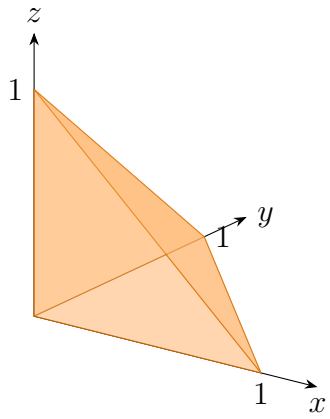
The same conclusion holds for any of the other five orders of integration.

More generally, suppose $\mathcal{W}$ lies over a domain $\mathcal{D}$ in the xy-plane and is bounded below and above by $z = z_1(x, y)$ and $z = z_2(x, y)$, respectively. Then

$$\iiint_{\mathcal{W}} f(x, y, z)\, \mathrm{d}V = \iint_{\mathcal{D}} \left(\int_{z_1(x,y)}^{z_2(x,y)} f(x, y, z)\, \mathrm{d}z \right) \mathrm{d}A.$$

Example 12.1 (Volume of a Tetrahedron)**.**

The plane $x + y + z = 1$ cuts off a tetrahedron $\mathcal{W}$ in the first octant. Describe $\mathcal{W}$ using inequalities in rectangular coordinates, then find its volume.



Solution 12.1

Because $\mathcal{W}$ lies in the first octant and below the plane $z = 1 - x - y$,

$$\mathcal{W} = \{(x, y, z) : x \geq 0,\ y \geq 0,\ z \geq 0,\ x + y + z \leq 1\}.$$

Using the order $\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x$, the bounds are

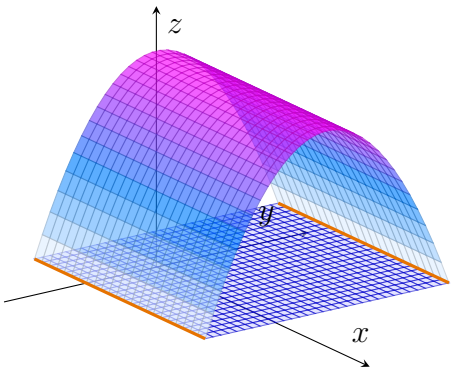
$$0 \leq x \leq 1, \qquad 0 \leq y \leq 1 - x, \qquad 0 \leq z \leq 1 - x - y.$$

Therefore,

$$\begin{aligned}
\mathrm{Vol}(\mathcal{W}) &= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} 1 \,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x \\
&= \int_0^1 \int_0^{1-x} (1 - x - y)\,\mathrm{d}y\,\mathrm{d}x \\
&= \frac{1}{2}\int_0^1 (1 - x)^2\,\mathrm{d}x = \frac{1}{6}.
\end{aligned}$$

Example 12.2 (Mass Between Two Surfaces)**.**

Let $\mathcal{W}$ be the solid bounded by $z = 1$, $z = 2 - y^2$, $x = 0$, and $x = 1$. The density at (x, y, z) is $f(x, y, z)$ kilograms per cubic meter. Write an iterated integral for the mass of $\mathcal{W}$.



Solution 12.2

The two z-surfaces intersect when

$$1 = 2 - y^2 \iff y = \pm 1.$$

Thus

$$\mathcal{W} = \{(x, y, z) : 0 \le x \le 1,\ -1 \le y \le 1,\ 1 \le z \le 2 - y^2\}.$$

The mass is

$$M = \int_0^1 \int_{-1}^1 \int_1^{2-y^2} f(x, y, z)\, \mathrm{d}z\, \mathrm{d}y\, \mathrm{d}x.$$

Example 12.3 (Choosing an Order of Integration)**.**

Let $\mathcal{W}$ be the solid in the first octant bounded by

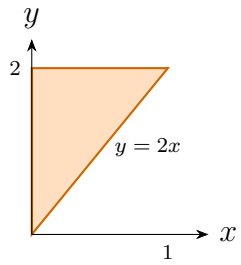
$$z = 4 - y^2, \qquad y = 2x, \qquad z = 0, \qquad x = 0.$$

Calculate

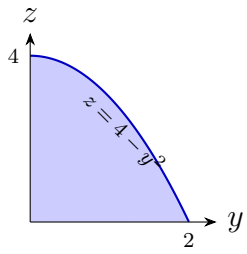
$$\iiint_{\mathcal{W}} xyz\, \mathrm{d}V.$$

The three coordinate-plane projections of $\mathcal{W}$ are shown below.

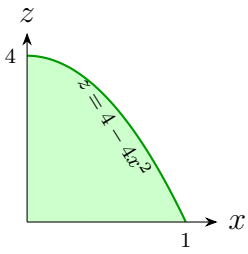
Projection onto the xy-plane



Projection onto the yz-plane



Projection onto the xz-plane



Solution 12.3

The projection onto the xy-plane is

$$D_{xy} = \{(x, y) : 0 \le y \le 2,\ 0 \le x \le \tfrac{y}{2}\}.$$

For each $(x, y) \in D_{xy}$, the variable z ranges from the xy-plane to the parabolic cylinder:

$$0 \le z \le 4 - y^2.$$

Therefore,

$$\begin{aligned}\iiint_{\mathcal{W}} xyz \,\mathrm{d}V &= \int_0^2 \int_0^{y/2} \int_0^{4-y^2} xyz \,\mathrm{d}z \,\mathrm{d}x \,\mathrm{d}y \\ &= \int_0^2 \int_0^{y/2} \frac{xy}{2}(4-y^2)^2 \,\mathrm{d}x \,\mathrm{d}y \\ &= \frac{1}{16}\int_0^2 y^3(4-y^2)^2 \,\mathrm{d}y.\end{aligned}$$

With $u = y^2$, so that $y^3 \,\mathrm{d}y = \frac{u}{2}\,\mathrm{d}u$,

$$\begin{aligned}\iiint_{\mathcal{W}} xyz \,\mathrm{d}V &= \frac{1}{32}\int_0^4 u(4-u)^2 \,\mathrm{d}u \\ &= \frac{2}{3}.\end{aligned}$$